End-to-End House Price Prediction

Exploratory Data Analysis • Feature Engineering • Regression • Model Evaluation • Gradio

Dataset: Kaggle House Prices: Advanced Regression Techniques (Ames, Iowa). This report uses the actual labelled `train.csv` supplied for the project. Prices are historical USD, not current market values.

1. Problem statement

Build a supervised machine-learning regression system that estimates historical house sale prices from property characteristics such as size, rooms, quality, condition, construction age, garage, basement, and neighborhood. The final model is exposed through a Gradio interface.

2. Dataset understanding

Dataset	Rows	Columns	Purpose
train.csv	1,460	81	Labelled training data; includes SalePrice
test.csv	1,459	80	Unlabelled records for final prediction
sample_submission.csv	1,459	2	Output format template; prices are placeholders

After feature engineering, the model uses **89** predictors. The labelled training data is split into 1,168 training rows and 292 validation rows for model evaluation.

`SalePrice` mean: **$180,921**; median: **$163,000**; minimum: **$34,900**; maximum: **$755,000**.

3. Data cleaning and feature engineering

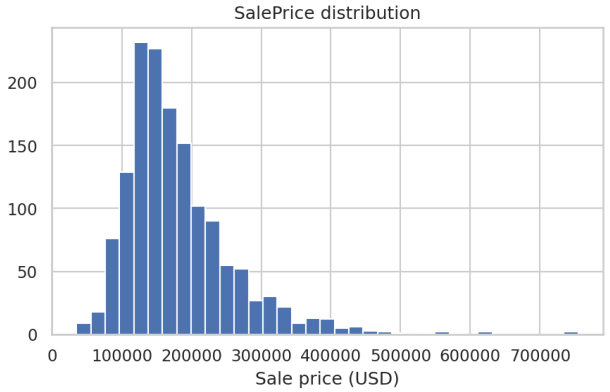
The `Id` field is excluded because it is only an identifier. `MSSubClass` is treated as a categorical dwelling-type code. Missing values are interpreted using the data dictionary: missing basement, garage, fireplace, pool, fence, or alley categories can mean that the feature is absent. Relevant absent-feature measurements are set to zero; other numerical gaps are imputed with the training median, and remaining categorical gaps are filled within the training pipeline.

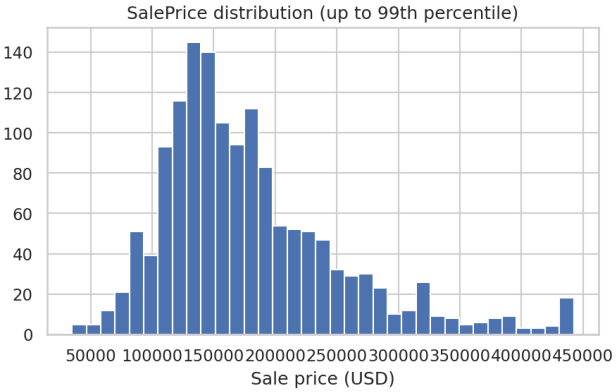
Engineered features include HouseAge, RemodAge, GarageAge, TotalArea, TotalBathrooms, TotalPorchSF, and indicators such as HasGarage and HasBasement. `PricePerSqFt` is not used as a model input because it is calculated using `SalePrice` and would cause target leakage.

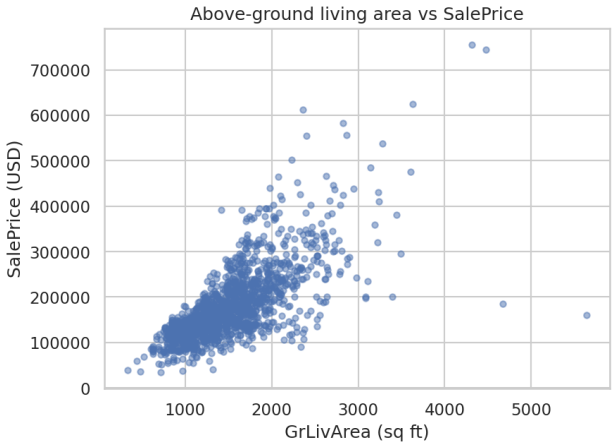
Numeric values are median-imputed and standardized. Categorical values are imputed and one-hot encoded. Preprocessing is fitted inside the cross-validation pipeline to reduce data leakage. The target is log-transformed during fitting and predictions are converted back to USD.

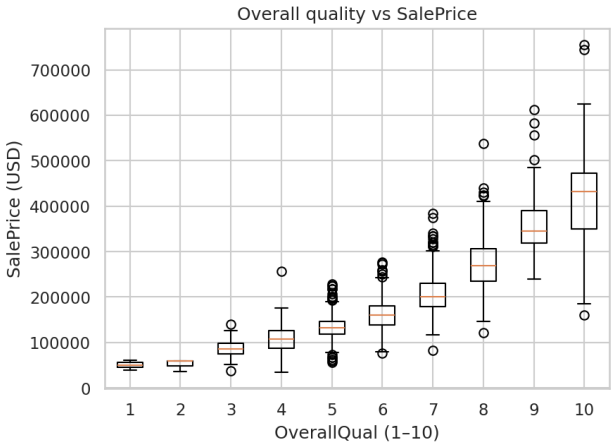
4. Exploratory data analysis

The charts below show target distribution, area and quality relationships, neighborhood price differences, missing values, and numeric correlations. Correlation and group averages describe associations; they do not prove causation.

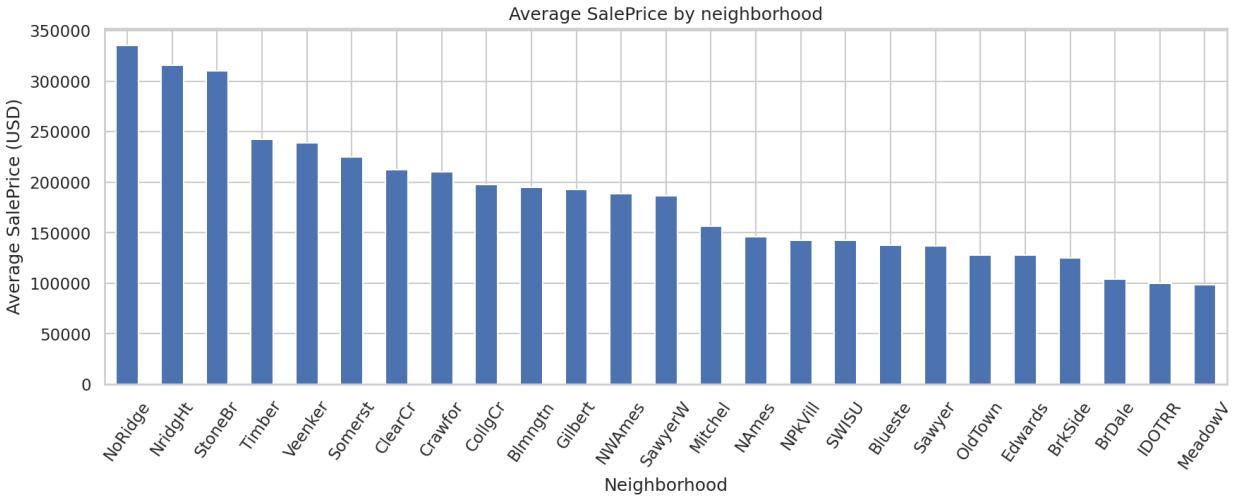


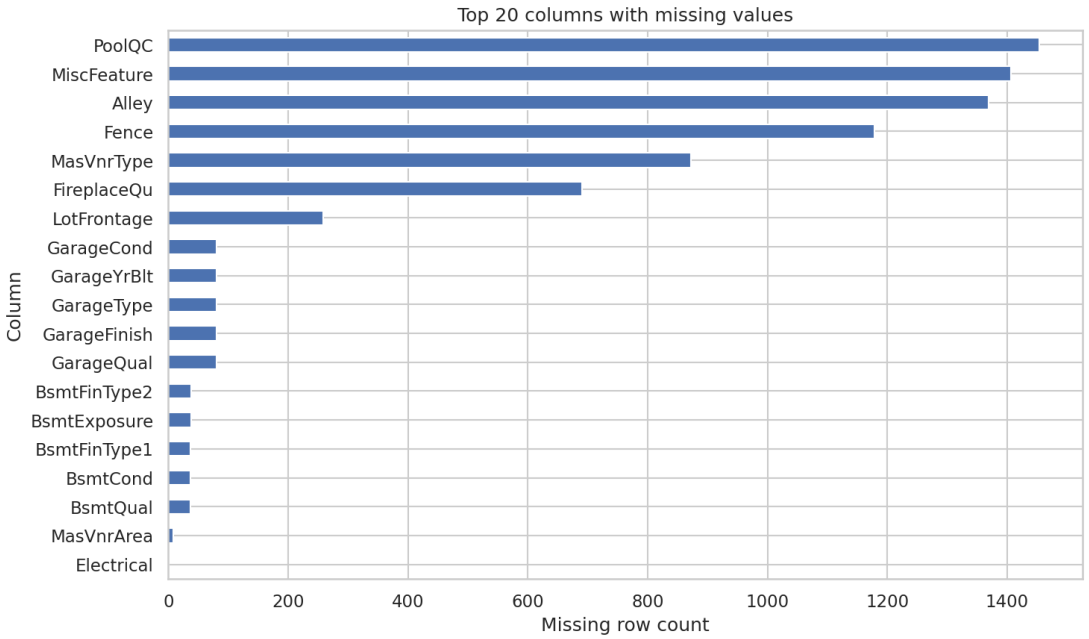


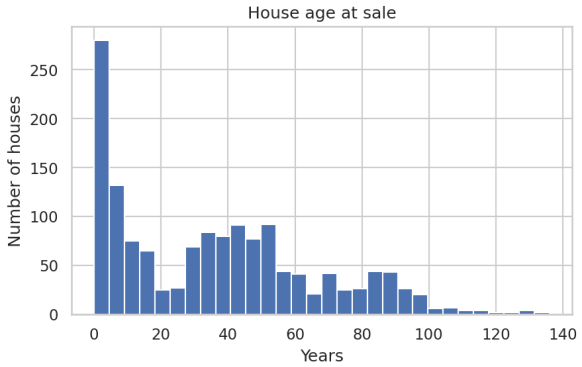

Above-ground living area vs SalePrice
SalePrice (USD)
0
100000
200000
300000
400000
500000
600000
700000
1000
2000
3000
4000
5000
GrLivArea (sq ft)

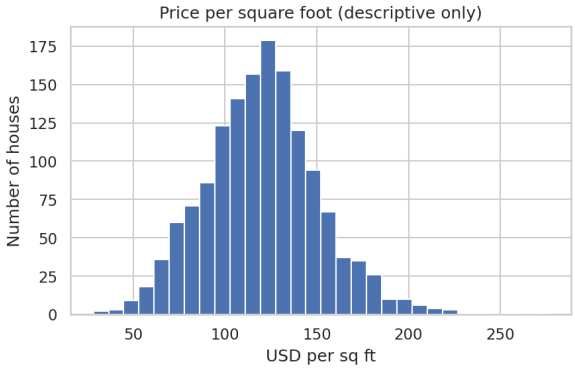

Overall quality vs SalePrice
SalePrice (USD)
0
100000
200000
300000
400000
500000
600000
700000
1
2
3
4
5
6
7
8
9
10
OverallQual (1–10)

5. Neighborhood and correlation findings



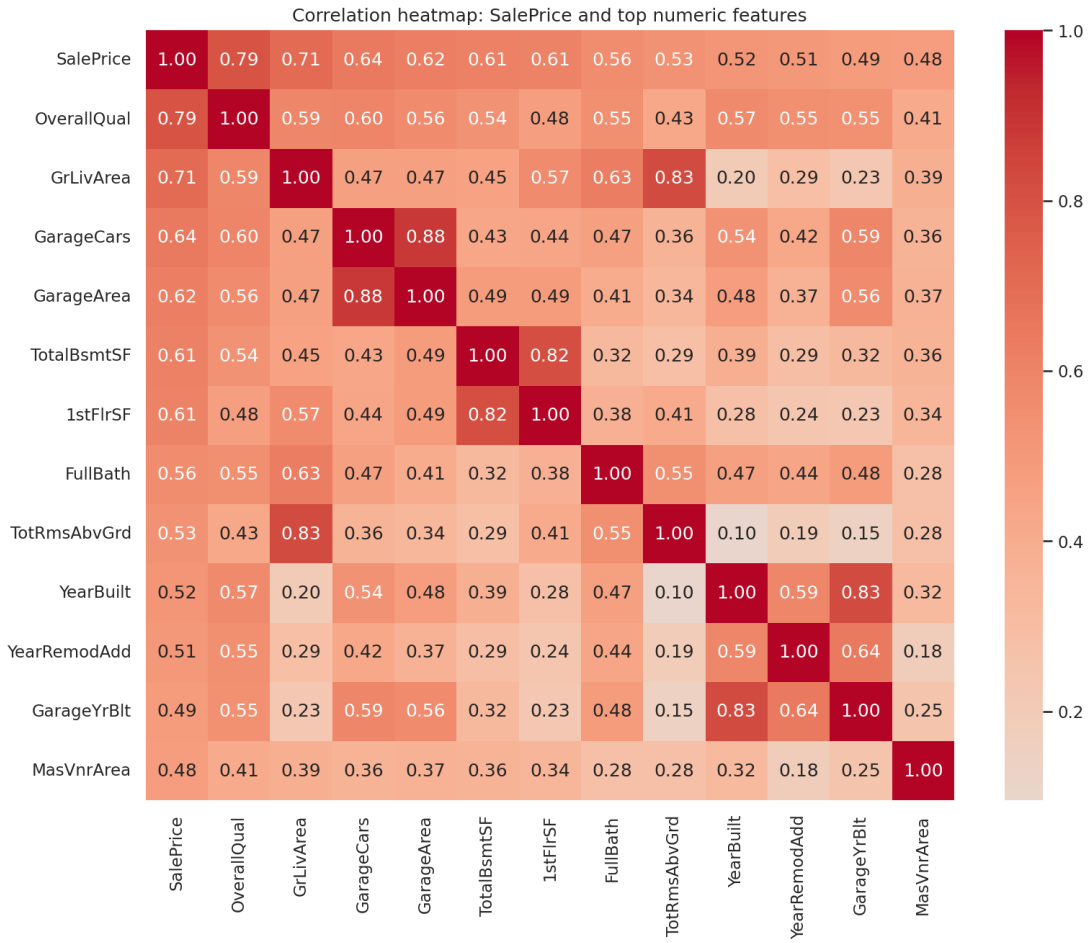






Numeric feature	Correlation with SalePrice
OverallQual	+0.791
GrLivArea	+0.709
GarageCars	+0.640
GarageArea	+0.623
TotalBsmtSF	+0.614

Numeric feature	Correlation with SalePrice
1stFlrSF	+0.606
FullBath	+0.561
TotRmsAbvGrd	+0.534



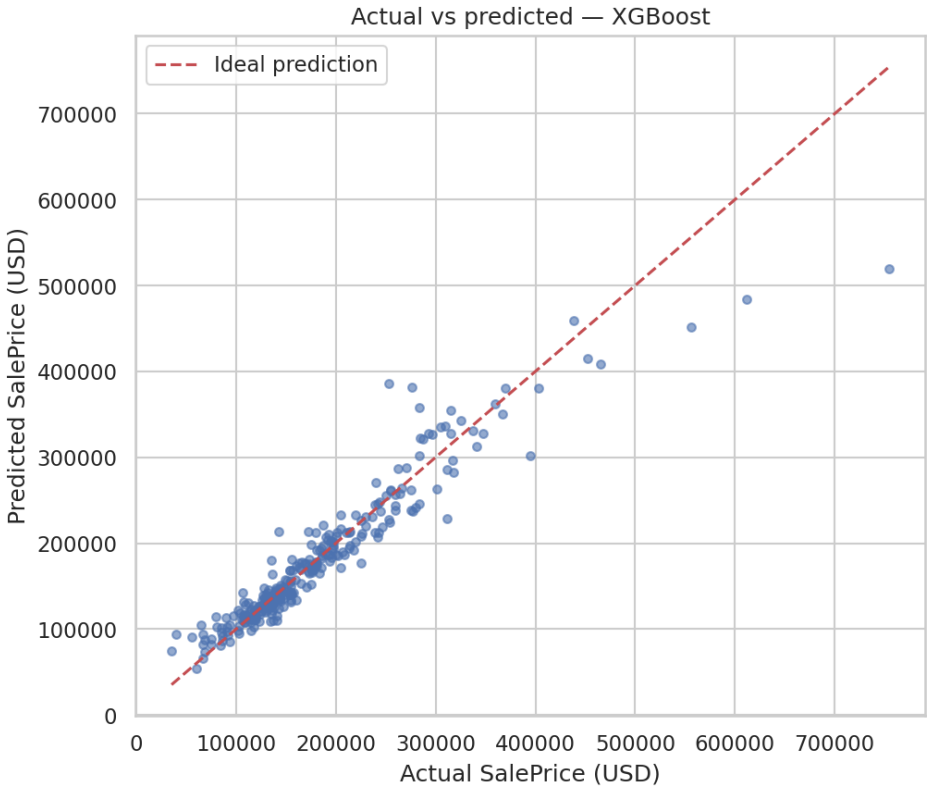
6. Model development and evaluation

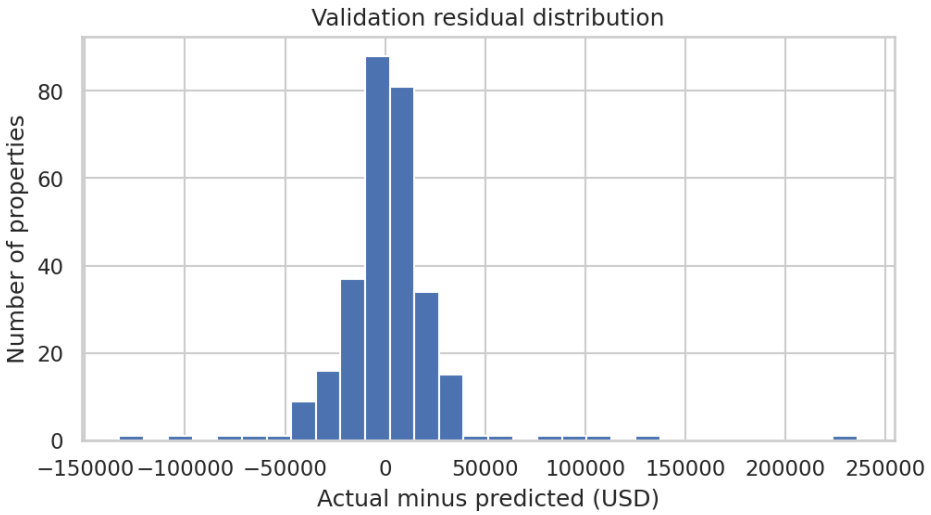
The labelled data is split using an 80/20 train-validation split (random_state=42). Linear Regression is used as a baseline. Random Forest and XGBoost are tuned using randomized hyperparameter search with 3-fold cross-validation. The selected model is chosen using cross-validation RMSE from the training portion; the validation set is retained for final evaluation.

Model	CV RMSE	Validation MAE	Validation RMSE	Validation R^2
XGBoost	$28,458	$15,896	$27,455	0.902
Random Forest	$30,806	$17,181	$30,312	0.880
Linear Regression	$58,253	$15,655	$24,294	0.923

Selected model by cross-validation: **XGBoost**. The validation metrics are reported separately and should be interpreted as an estimate on a held-out subset of this historical dataset.

On this particular holdout split, Linear Regression has a lower validation RMSE than XGBoost, even though XGBoost has the lowest average cross-validation RMSE. This difference illustrates that model rankings can vary across data splits; cross-validation is used for model selection and the holdout metrics are reported transparently as a separate final check.





7. Kaggle test prediction

The selected model was refitted using all 1,460 labelled training rows and used to predict the 1,459 records in `test.csv`. The generated `submission.csv` contains `Id` and predicted `SalePrice`, matching the structure of `sample_submission.csv`. Since test.csv has no actual SalePrice values, its accuracy cannot be calculated from the supplied files.

8. Business recommendations

Historical data can help a real-estate team understand how living area, overall quality, neighborhood, basement, garage, and age relate to sale prices. Use estimates as decision support, not as a replacement for professional appraisal. A real deployment should use recent verified transactions from the target city, show uncertainty intervals, check errors by neighborhood and price range, and retrain as market conditions change.

9. Limitations and responsible interpretation

The Ames dataset represents historical sales in Ames, Iowa, primarily 2006–2010. Prices are in historical USD. The model is not trained on Nagpur or Indian real-estate data and must not be interpreted as a current property valuation in India. The Gradio form exposes key fields and fills other fields with representative training-data values, so its predictions are illustrative. This is an

educational project, not a certified valuation.

10. Deliverables and reproducibility

Project deliverables include two Jupyter notebooks, reusable Python modules, a saved preprocessing/model pipeline (`artifacts/best_house_price_model.pkl`), model comparison (`artifacts/model_comparison.csv`), EDA/evaluation charts, a Gradio interface (`app/app.py`), and `submission.csv`.

From the project root: install packages with `pip install -r requirements.txt`; run `python train_model.py`; generate this report with `python generate_report.py`; launch the interface with `python app/app.py`.

11. References

Kaggle — House Prices: Advanced Regression Techniques: https://www.kaggle.com/competitions/house-prices-advanced-regression-techniques/data

L&T; Fake News Classification reference repository (project organization only): https://github.com/akashkeote/L-and-T-Assignment-Fake-News-Classification